

SAT Math Mastery · Week 6B

Center, Spread & Plots

Time	150 minutes: 110 min solving + 40 min self-check and error log
Topics	mean, median, mode, range · standard deviation · adding, removing and changing a value · weighted means · frequency tables · box plots, dot plots, histograms · comparing spreads without computing
Name / Date	_____

1 · SKILLS & IDEAS

- S1.** Compute mean, median, mode and range from a list, a frequency table or a dot plot.
- S2.** Use the total: $\text{sum} = n \times \text{mean}$; find a missing value that gives a target mean.
- S3.** Predict the effect of adding, removing or changing one value on the mean and on the median position.
- S4.** Combine groups with a weighted mean; recover a group size from the combined mean.
- S5.** Read the median from running totals (frequency tables, dot plots, histograms with bins).
- S6.** Read a box plot: five-number summary, IQR, range; know what a box plot cannot show (the mean).
- S7.** Compare standard deviations by distance from the mean, without computing.
- S8.** Apply $y = ax + b$ to a data set: effect on mean, median, range, IQR and standard deviation.
- S9.** Bound the possible mean or median from a histogram or a partly known data set.

2 · REQUIRED FACTS

- Mean = $\frac{\text{sum}}{n}$. Median = middle value of the ordered list (average of the two middle values when n is even). Mode = most frequent value. Range = $\text{max} - \text{min}$.
- The median is found by position, so changing an extreme value does not move it; the mean does move.
- Adding a value equal to the mean leaves the mean unchanged and *reduces* the standard deviation.
- Standard deviation measures typical distance from the mean: values farther from the mean raise it, values closer lower it.
- Adding a constant to every value shifts mean and median but leaves range, IQR and SD unchanged.
- A box plot shows min, Q_1 , median, Q_3 , max; each of the four sections holds about 25% of the data. It does not show the mean.

3 · REQUIRED FORMULAS

$$\bar{x} = \frac{\sum x}{n}, \quad \bar{x}_{\text{new}} = \frac{n\bar{x} + v}{n+1}, \quad \bar{x}_{\text{combined}} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}, \quad \text{median position} = \frac{n+1}{2}$$

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}, \quad \text{IQR} = Q_3 - Q_1, \quad \text{mean from a table} = \frac{\sum fx}{\sum f}$$

$$y = ax + b: \quad \bar{y} = a\bar{x} + b, \quad \text{median}_y = a \text{median}_x + b, \quad \sigma_y = |a|\sigma_x, \quad \text{range}_y = |a|\text{range}_x$$

4 · MINI THEORY SHEET

Which measure answers which question.

Question	Tool
a missing value for a target mean	target sum – known sum
effect of one new value v	new mean = $\frac{n\bar{x}+v}{n+1}$; the median moves at most one position
two groups together	weighted mean (weights are the group sizes)
median from a frequency table	running total until it passes $\frac{n+1}{2}$ (or the two middle positions)
which set is more spread out	compare distances from the mean, not the range
a histogram's mean	least: every value at the left edge; greatest: right edge

Box plot. The ends of the box are Q_1 and Q_3 , the line inside is the median, the whiskers reach min and max. Range = max – min; IQR = box length.

Dot plot and histogram. Each dot is one value, so count dots in order to reach the middle position. In a histogram, bar heights are frequencies; the total is the sum of the heights. The median lies in the first bar whose running total reaches the middle position(s).

Transformations. Multiplying all values by a scales the spread by $|a|$; adding b shifts the center by b and does not change the spread.

Speed shortcuts.

Situation	Shortcut
add a value to reach a target mean	needed value = target mean + (target mean – old mean) $\times n$
replace one value u by w	mean changes by $\frac{w-u}{n}$; the median is unchanged unless the middle moves
add c to every value	mean and median shift by c ; range, IQR, SD unchanged
two groups, combined mean	the combined mean sits closer to the larger group
SD without computing	mostly-far-from-mean beats a long thin tail of near values
histogram mean bounds	use left edges for the least, right edges for the greatest
box plot counts	each quarter holds 25% of the values

5 · GUIDED EXAMPLES

Example 1 — Total, new mean and median

The mean of 6 values is 14. A seventh value, 28, is added. Find the new mean. Then find the median of 2, 5, 5, 9, 12, 20.

Total of the six: $6(14) = 84$. New mean = $\frac{84+28}{7} = \frac{112}{7} = 16$.

Six values (even n): median is the average of the 3rd and 4th: $\frac{5+9}{2} = 7$.

Example 2 — Frequency table: mean and median

Scores 1, 2, 3, 4 occur 3, 5, 4, 2 times. Find the mean and the median.

$n = 3 + 5 + 4 + 2 = 14$; $\sum fx = 3(1) + 5(2) + 4(3) + 2(4) = 33$, so the mean is $\frac{33}{14} \approx 2.36$.

Median: average of the 7th and 8th values. Running totals 3, 8, ... show both are in the score-2 group, so the median is 2.

Example 3 — Spread under transformations

Set $P = \{3, 5, 7, 9, 11\}$ has mean 7, range 8, and $\sigma = \sqrt{\frac{16+4+0+4+16}{5}} = \sqrt{8}$. Set Q is every value of P plus 10; set R is every value of P times 2.

Q : mean 17, range 8, $\sigma = \sqrt{8}$ (shift: spread unchanged).

R : mean 14, range 16, $\sigma = 2\sqrt{8}$ (scaling: spread doubles).

6 · PRACTICE SET — 37 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. EASY · NO CALCULATOR NEEDED

The mean of five numbers is 17. Four of the numbers are 12, 20, 15, and 25. What is the fifth number?

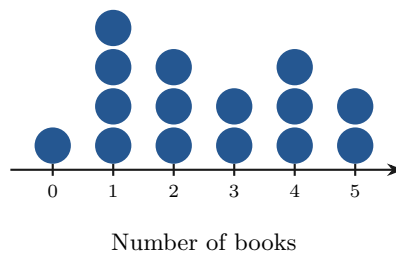
2. EASY · NO CALCULATOR NEEDED

What is the median of the data set 14, 9, 22, 17, 9, 31, 26, 12?

- (A) 9 (B) 15.5
(C) 17 (D) 17.5

3. EASY · NO CALCULATOR NEEDED

The dot plot shows the number of books each of 15 students read last month. What is the median number of books read?



- (A) 3 (B) 1
(C) 2 (D) 2.5

4. EASY · NO CALCULATOR NEEDED
A data set of 6 values has a mean of 20. A seventh value, 34, is added to the data set. What is the mean of the 7 values?

- (A) 22
- (B) 24
- (C) 27
- (D) 21
-
-
-
-

5. EASY · CALCULATOR
The table shows the number of pets owned by each student in a class. What is the mean number of pets per student?

Pets	0	1	2	3
Students	7	11	8	4

6. MEDIUM · NO CALCULATOR NEEDED
The table shows the scores of 30 students on a quiz. What is the median score?

Score	1	2	3	4	5
Frequency	5	6	4	9	6

- (A) 4
- (B) 3.2
- (C) 3
- (D) 3.5
-
-
-
-

7.

MEDIUM · NO CALCULATOR NEEDED

The mean of 7 test scores is 78. When one score is removed, the mean of the remaining 6 scores is 80. What was the score that was removed?

8.

MEDIUM · NO CALCULATOR NEEDED

Class A has 24 students with a mean test score of 82. Class B has 16 students with a mean test score of 72. What is the mean test score of all 40 students?

- (A) 80 (B) 77
(C) 78 (D) 79

9.

MEDIUM · NO CALCULATOR NEEDED

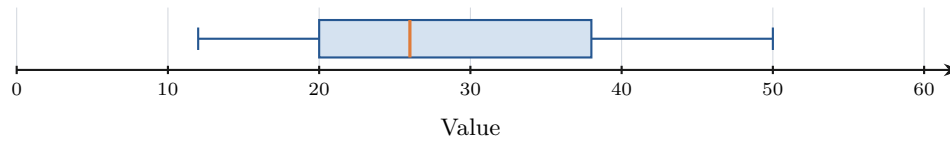
A data set has a mean of 40, a median of 38, a standard deviation of 6, and a range of 25. Every value in the data set is multiplied by 2, and then 3 is subtracted from each result. What is the standard deviation of the new data set?

- (A) 9 (B) 12
(C) 15 (D) 24

10.

MEDIUM · NO CALCULATOR NEEDED

The box plot summarizes a data set. Which of the following must be true about the data set?



(A) The standard deviation of the data set is 18.

(B) Exactly half of the data values are between 12 and 26.

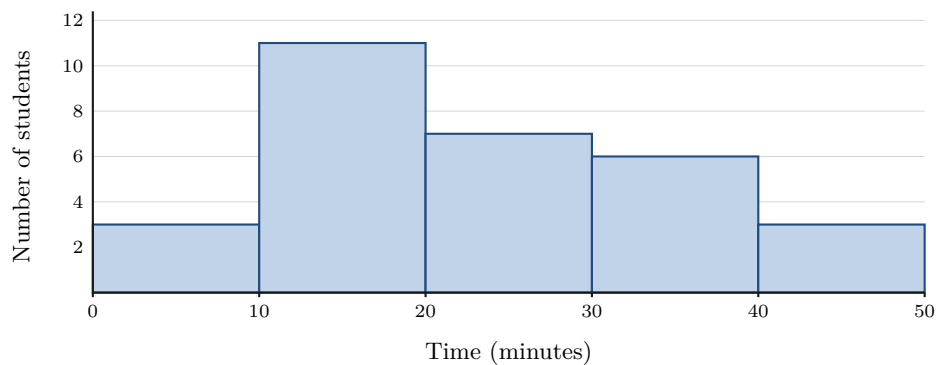
(C) The mean of the data set is 26.

(D) At least 25% of the data values are greater than or equal to 38.

11.

MEDIUM · NO CALCULATOR NEEDED

The histogram shows the distribution of the times, in minutes, that 30 students took to finish a puzzle. In which interval does the median time lie?



(A) 20 to 30

(B) 30 to 40

(C) 0 to 10

(D) 10 to 20

12.

MEDIUM · NO CALCULATOR NEEDED

The data set 6, 9, 11, 14, 20 has a median of 11. A sixth value x is added to the data set, and the median of the new six-value data set is 12.5. Which of the following could be the value of x ?

- (A) 13 (B) 16
(C) 8 (D) 12

13.

MEDIUM · NO CALCULATOR NEEDED

Maria's scores on her first five quizzes are 78, 91, 84, 88, and 80. What score must she earn on her sixth quiz so that the mean of all six scores is exactly 85?

14.

HARD · NO CALCULATOR NEEDED

Group A has 30 members with a mean score of 70. Group B has a mean score of 90. The mean score of all the members of both groups combined is 78. How many members are in Group B?

15.

HARD · NO CALCULATOR NEEDED

The mean of 12 values is 45. A new value is added, and the mean of all 13 values is 47. What is the value that was added?

16.

HARD · NO CALCULATOR NEEDED

Data set A is 0, 10, 10, 10, 10, 10, 20. Data set B is 3, 3, 5, 10, 15, 17, 17. Both data sets have a mean of 10. Which of the following statements about the standard deviations of the two data sets is true?

(A) Data set A has the greater standard deviation because more of its values equal the mean.

(B) Data set A has the greater standard deviation because its range is greater.

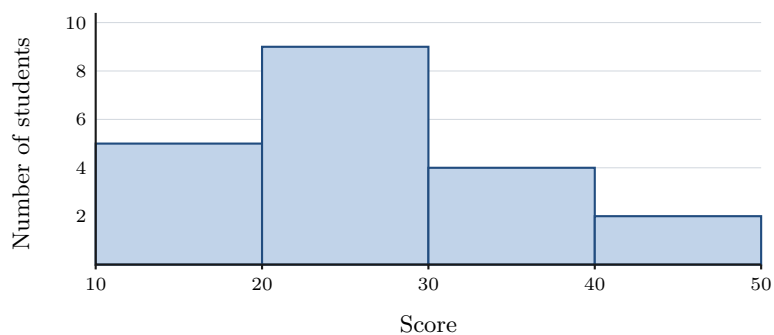
(C) Data set B has the greater standard deviation because its values are generally farther from the mean.

(D) The two standard deviations are equal because the means are equal.

17.

HARD · NO CALCULATOR NEEDED

The histogram shows the distribution of scores for 20 students on a test. Each interval includes its left endpoint but not its right endpoint. Which of the following could be the mean score of the 20 students?



(A) 24

(B) 33

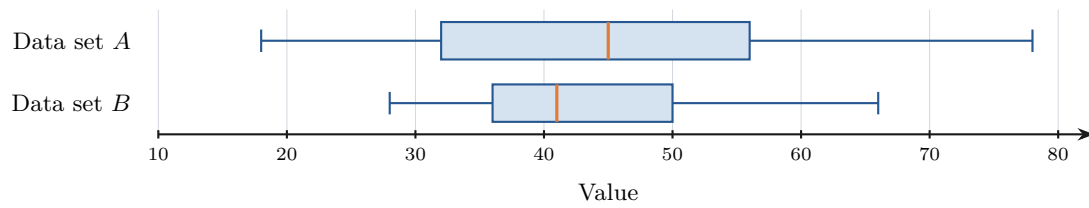
(C) 38

(D) 19

18.

HARD · NO CALCULATOR NEEDED

The box plots summarize two data sets, A and B . Which of the following statements must be true?



(A) More than 75% of the values in data set B are less than 50.

(B) The range of data set B is greater than the range of data set A .

(C) The mean of data set A is greater than the mean of data set B .

(D) At least 25% of the values in data set A are greater than 50.

19.

HARD · NO CALCULATOR NEEDED

The mean of the five numbers x , $x + 4$, $2x - 1$, $3x + 2$, and $x - 9$ is 20. What is the median of the five numbers?

20.

HARD · NO CALCULATOR NEEDED

Five distinct positive integers have a mean of 20 and a median of 18. What is the greatest possible value of the largest of the five integers?

21.

HARD · NO CALCULATOR NEEDED

A data set of 9 values has a mean of 52 and a median of 50. The greatest value in the data set, 90, is replaced by 108. Which of the following describes the effect on the mean and median?

- (A) The mean increases by 2 and the median increases by 18. (B) The mean and the median both increase by 2.
- (C) The mean increases by 2 and the median is unchanged. (D) The mean increases by 18 and the median is unchanged.

22.

HARD · NO CALCULATOR NEEDED

The table shows the number of siblings of each of 15 students.

Siblings	0	1	2	3	4
Students	3	5	4	2	1

Four more students are surveyed, and each of them has more siblings than any of the original 15 students. What is the median number of siblings for all 19 students?

- (A) 1.5 (B) 2
(C) 3 (D) 1

23.

HARD · NO CALCULATOR NEEDED

A data set has a mean of 30 and a standard deviation of 4. Each value x in the data set is replaced by $y = ax + b$, where $a > 0$. The new data set has a mean of 95 and a standard deviation of 10. What is the value of b ?

- (A) 65 (B) 75
(C) 6 (D) 20

24.

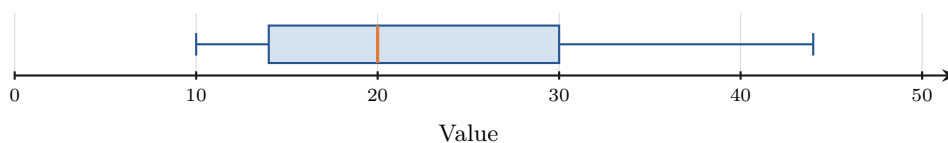
SAT CHALLENGE • NO CALCULATOR NEEDED

The data set 2, 6, k , 10, $3k$ consists of five numbers, where k is a positive integer. The mean of the data set equals its median. What is the value of k ?

25.

SAT CHALLENGE • NO CALCULATOR NEEDED

The box plot summarizes a data set of 11 distinct whole numbers. For these 11 values, Q_1 is the median of the lowest five values and Q_3 is the median of the highest five values. What is the greatest possible sum of the 11 values?



26.

SAT CHALLENGE · NO CALCULATOR NEEDED

Five distinct positive integers have a mean of 10 and a median of 9. Which of the following could *not* be the greatest of the five integers?

(A) 13 (B) 17

(C) 25 (D) 28

27.

SAT CHALLENGE · NO CALCULATOR NEEDED

Data set A is 2, 4, 6, 8, 10. Data set B consists of the values of A together with one additional value, 6. Data set C consists of the values of A together with the two additional values 0 and 12. Data set D consists of the values of A together with the two additional values 5 and 7. Consider the following statements about standard deviations.

I. The standard deviation of B is less than that of A .

II. The standard deviation of C is greater than that of A .

III. The standard deviation of D is greater than that of B .

Which of the statements must be true?

(A) I only (B) I and II only

(C) II and III only (D) I, II, and III

28.

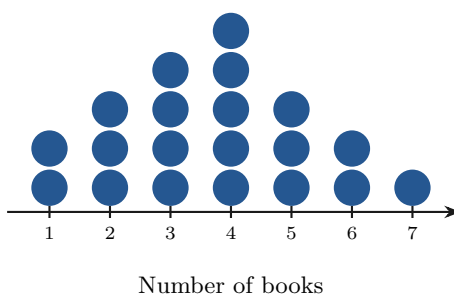
SAT CHALLENGE · NO CALCULATOR NEEDED

Group A has a mean score of 72 and Group B has a mean score of 84. The mean score of all 45 people in the two groups combined is 80. A third group, C , of 15 people is then added, and the mean score of all 60 people in the three groups is 79. What is the mean score of Group C ?

29.

SAT CHALLENGE · NO CALCULATOR NEEDED

The dot plot shows the number of books read last month by each of 20 students. Then k more students are surveyed, and each of them read 8 books. For how many positive integer values of k is the median of all $20 + k$ values still equal to the median of the original 20 values, while the mean of all $20 + k$ values is at least 4.5?



30.

SAT CHALLENGE · NO CALCULATOR NEEDED

The table shows the scores of 30 students on a quiz; the frequencies of scores 2 and 4 are a and b . The mean score is 3.2. What is the median score of the 30 students?

Score	1	2	3	4	5
Frequency	4	a	7	b	3

31.

SAT CHALLENGE · NO CALCULATOR NEEDED

A data set has eight values: 12, 15, 18, 20, 20, 22, 25, 28. The values 12 and 28 are removed and the values 19 and 21 are added in their place. Which of the following describes the mean and the standard deviation (SD) of the new data set compared with the original data set?

(A) Mean smaller, SD smaller

(B) Mean same, SD larger

(C) Mean same, SD smaller

(D) Mean same, SD same

32.

SAT CHALLENGE · CALCULATOR

A data set consists of the five values 5, 8, 17, x , x , where x is a real number. The variance of a data set is the mean of the squared deviations of its values from their mean. What is the least possible value of the variance of this data set?

- (A) 15.6 (B) 10
(C) 26 (D) 78

33.

SAT CHALLENGE · CALCULATOR

A data set has 10 values: 31, 38, 42, 45, 47, 52, 53, 58, 65, 69. The mean of the data set is 50. The least value and the greatest value are removed, leaving 8 values. Which of the following is true of the new data set compared with the original? (SD means standard deviation.)

- (A) Mean unchanged; SD unchanged (B) Mean smaller; SD smaller
(C) Mean greater; SD smaller (D) Mean unchanged; SD smaller

34.

SAT CHALLENGE · NO CALCULATOR NEEDED

A list of nine positive integers has a mean of 12 and a median of 9. What is the greatest possible value of the sum of the three largest integers in the list?

35.

SAT CHALLENGE · NO CALCULATOR NEEDED

Nine positive integers, not necessarily distinct, have a sum of 100 and a median of 9. What is the greatest possible value of the mean of the four largest of these integers?

(A) 25

(B) 20.25

(C) 21.75

(D) 23.75

36.

SAT CHALLENGE · NO CALCULATOR NEEDED

A data set has six values: 2, 3, 5, 12, 18, 20. Every value is increased by 4, and then a seventh value equal to the mean of the six increased values is added to the data set. Consider the following statements about the new data set of seven values compared with the original data set.

- I. The mean is 4 greater.
- II. The standard deviation is smaller.
- III. The median is 4 greater.

Which of the statements must be true?

- (A) I only
- (B) I and II only
- (C) II and III only
- (D) I, II and III

37.

SAT CHALLENGE · NO CALCULATOR NEEDED

A data set has a median of 52 and an interquartile range (IQR) of 14. Each value x in the data set is replaced by $15 - \frac{x}{2}$. What is the sum of the median and the IQR of the new data set?

- (A) 3
- (B) 11
- (C) -18
- (D) -4